

Ph.D. Pre – Viva Presentation

Assessment of the Effect of Global Warming on Phytoplankton Biomass in Coastal Waters of Tanzania

Venue: Kunduchi Boardroom

22/05/2023

Presenter: Peter Nyamisi

Time: 8:50 – 10: 40

Comments from Opponents

This study provides valuable insights into the effects of global warming on phytoplankton biomass in the coastal waters of Tanzania. The use of satellite data and air temperature data allows for a comprehensive assessment of the impacts of global warming on phytoplankton. The findings suggest that global warming has had a significant impact on phytoplankton biomass, with warmer temperatures leading to an increase in phytoplankton biomass. However, the thesis needs major revision before the next step.

General comments

1. The data collected from 2003 to 2020 do still represents the current situation in the WIO region?
2. One of the potential limitations of this study is the relatively short time frame analyzed (2003–2020), which may not fully capture long-term trends and fluctuations in phytoplankton biomass. Additionally, it would be valuable to consider the impacts of other environmental factors, such as ocean acidification, on phytoplankton growth and distribution.
3. Overall, the paper is not well structured and contains many spelling, grammar, and number-heading issues that need to be addressed. Checking for spelling and grammar errors in the entire thesis
4. In the model, why did you choose the period 2013–2020 as your base or reference period?
5. Which strategic plan do you think these findings will fit into? Which policy can be improved based on this study?
6. What was the main issue that led to your study, and how did you solve it?
7. How will your research findings benefit the community, in layman's language?
8. What is the key take-home message? The new contribution of your study to knowledge
9. What are the implications of the key findings?

Abstract

1. The abstract lacks the concluding statement(s) based on the overall aim of the study.

2.The findings have to be synthesized, or the implications of the key findings are missing.

3.The first sentence says, "The effects of global warming on phytoplankton biomass estimated using chlorophyll-a (Chl-a) concentration was assessed along the coastal waters in Tanzania". Why this assessment? Is there a problem or gap in Tanzania?

Introduction

1.In the section you quote Semba et al. (2016, I am not sure if the statement is true because, according to the author, the temperature trend or increase was insignificant. (See the abstract: "While in your study the phytoplankton displayed a significant declining trend, sea surface temperature showed an insignificant increasing trend".

2.The introduction lacks the aim of the study. Kindly Introduce the aim of the study.

Literature review;

1. In this section, it could be healthier to have a sub-section covering global warming indices that are more likely applicable to this study.
2. It could be wise to have a section or a standalone paragraph that shows how phytoplankton influences climate processes and biogeochemical cycles, particularly the carbon cycle.
3. Some of your paragraphs are constituted by two sentences or very long sentences; I advise rephrasing the sentences. How does a paragraph with two sentences work?
4. The literature review is well written, but it is only concentrated on phytoplankton. You should also write about global warming. It suggests defining the term global warming for readers to carry the main message.
5. Section 1.2.4, On the effect of global warming, how do you differentiate between marine and oceanic environments?

Problem statement, hypothesis, or research question

1.The problem statement has to be improved, as it should comprise (i) what is already known. (ii) What do we still need to know? And relevance, or why do we need to know more about the problem?

2. In that context, the statement lacks "why we need to know more about the gap identified. Using the word "lack of to mean "causes" or "leads to, what are the consequences of the unavailability of these studies?

3. After objectives, hypotheses or research questions should be followed. Not indicated I wonder why you did not include the hypothesis in your study. I wonder why you did not involve the hypothesis in your study.

Significance of the study

1. Will the study only be used in the future? Or even at present for clarification of uncertainties? Does the study mostly focus on the future? In the present, does the study not have a significant impact?
2. I recommend adding clarification on how the findings and study can be applied to manage marine resources sustainably.

Material and methods

1. The study area and justification of the chosen sites are not described or presented, e.g., data for nutrients are missing, as are where and how to collect them. Unless otherwise stated, the format used and the statistical analysis methods not presented
2. Pemba Channel, Zanzibar Channel, Mafia Channel, and Mtwara Site Where are these sites located? There is a need to describe these sites and ecosystems and why they exist.
3. Section 2.1.6. In situ data collection for nutrients is too shallow; it lacks where, why, when, and how.
4. How did you collect your water samples for laboratory analysis of primary production? The data on external nutrient sources were not quantified in this study. Did you collect the sample when, where, in what sampling zone, etc.?
5. Section 2.2.2: "Analysis of these data was done following the hypothesis set". Where are the hypotheses set forth in your thesis?
6. In Section 3.2, there is a paragraph with only two sentences. Is this a standalone paragraph? How does a paragraph have two sentences? Cross-check the entire thesis and avoid very short paragraphs.
7. Which statistical test you used to test the data for normality?
- 8.

Results

1. In the results section, some statements should be placed in the discussion part. This sound is a part of the discussion.
2. Please the Justification on why Mtwara presented the highest rate of temperature increase should be provided. Also, the depth of the Mtwara site was not described as in the case of other sites.
3. Map section; Since the study considered some river discharges as one of the global warming indices, the map should show the main rivers considered and the justification for why these rivers and these study channels or sites were chosen
4. Fig. 4.3: Make sure the labels of figures c and d are visible.
5. Figure 4.4: Heatmap showing monthly and interannual variation in SST at the
6. Mafia Channel; b) Mtwara Site; c) Pemba Channel; and d) Zanzibar Channel. a,b, c, and d where they are in the figures
7. I think it is important to highlight the differences between the results reported in Section 4.3.2 and this one in Section 4.3.4 to avoid having the results presented redundantly.

Discussion section

1. Section 4.4 Apart from the literature to justify the SE monsoons exhibiting higher Chl-a, this should also be supported by the present research findings on the period or season of heavy rains. When were heavy rains noted, as far as the research results are concerned, during the NE or SE monsoons?
2. Section 4.4 Why such a pattern? Do the precipitation and discharge-related data in the present study (as you tried to justify) reflect the differences in trends between 2003 and 2010 and those of 2011 and 2020?
3. 4.4 The point of evaporation in the atmosphere and its influence on the phytoplankton biomass should be added to the discussion sub-section.
4. 5.4 "This will probably cause SCM of low chlorophyll-a fluorescence to occur at relatively higher depths." Unless you are 100% sure or otherwise, don't use terms that imply the issue is entirely the product of your findings. This is due to the fact that not all the contributing aspects or factors were taken into account or explored in your study. This should be observed in all discussion sections of the thesis.

Conclusion What is the implication of your findings?

References

Citation style is different. Cross check the entire thesis.

It seems all citations used in the text are not linked to the reference. Probably she didn't use any citation tool/ bibliographic management tool or because of the software she used to write the report. So, it is hard to link the in-text citation and the reference/bibliography

Questions/Comments from other members

1. The quality of the graphs was good. However, some of the figures are too small to be seen.
2. You didn't mention the criteria for selecting study sites (why these sites?). You need to clarify this.
3. There was no discussion in your presentation; why? You have to discuss your results with the audience. It's important to include the discussion section in your presentation.
4. What is the difference between discharge and precipitation?
5. In your discussion, you should also consider the effects of global warming in fisheries.
6. Why was the Kimbiji site suddenly added when it was not there?
7. On your map, you can think about indicating some major rivers.
8. Why Mtwara has high productivity, Chl a, etc.
9. Are there any other factors that control chlorophyll's productivity? If yes, how did you rule it out?

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